# The velocity–wall-normal-vorticity (Kim–Moin–Moser) formulation for the spectral channel-flow solver

Working derivation for review

September 9, 2026

This note records, step by step, the velocity / wall-normal-vorticity ("$v$–$\eta$") formulation used by the solver, in the convention of the SIMSON user guide. The aim is a self-contained derivation that can be checked line by line. Equation references of the form (4.x) point to the SIMSON guide.

*See also:* `ZERO_TO_VETA.md` maps the end-to-end algorithm and the routine-level call graph — how the equations derived here are actually stepped in the code.

## 1 Governing equations

We start from the non-dimensional incompressible Navier–Stokes equations in a (possibly) rotating frame, in tensor notation (SIMSON (4.1)–(4.2)):

$$\frac{\partial u_i}{\partial t} = -\frac{\partial p}{\partial x_i} + \varepsilon_{ijk}\, u_j\, (\omega_k + 2\Omega_k) - \frac{\partial}{\partial x_i}\big(\tfrac{1}{2}\, u_j u_j\big) + \frac{1}{\mathrm{Re}}\, \nabla^2 u_i + F_i, \tag{1}$$

$$\frac{\partial u_i}{\partial x_i} = 0. \tag{2}$$

Here $\boldsymbol{u} = (u, v, w)$ are the streamwise, wall-normal and spanwise velocity components along $(x, y, z) = (x_1, x_2, x_3)$, $\boldsymbol{\omega} = \nabla \times \boldsymbol{u}$ is the vorticity, $\boldsymbol{\Omega}$ the (constant) frame rotation, and $\boldsymbol{F}$ a body force. The flow is periodic in $x$ and $z$ and bounded by no-slip walls at $y = \pm 1$.

Following SIMSON, collect the rotational/forcing terms into

$$H_i \;=\; \varepsilon_{ijk}\, u_j\, (\omega_k + 2\Omega_k) + F_i, \tag{3}$$

i.e. $\boldsymbol{H} = \boldsymbol{u} \times (\boldsymbol{\omega} + 2\boldsymbol{\Omega}) + \boldsymbol{F}$. For the non-rotating channel $\boldsymbol{\Omega} = \boldsymbol{0}$ and the convective core is simply $\boldsymbol{u} \times \boldsymbol{\omega}$. Absorbing the kinetic-energy gradient into a modified pressure,

$$\Pi \;=\; p + \tfrac{1}{2}\, u_j u_j, \tag{4}$$

equation (1) becomes

$$\frac{\partial u_i}{\partial t} \;=\; -\frac{\partial \Pi}{\partial x_i} + H_i + \frac{1}{\mathrm{Re}}\, \nabla^2 u_i. \tag{5}$$

The strategy of the $v$–$\eta$ formulation is to apply differential operators to (5) that annihilate $\nabla\Pi$, producing closed evolution equations for $v$ and $\eta \equiv \omega_y$ in which pressure never appears.

## 2 The wall-normal velocity equation

Apply the operator $\big(\nabla \times \nabla \times \cdot\big)$ to (5) and take the $y$-component. We use the identity

$$\nabla \times \nabla \times \boldsymbol{a} \;=\; \nabla(\nabla \cdot \boldsymbol{a}) - \nabla^2 \boldsymbol{a}. \tag{6}$$

**Left-hand side.** With $\boldsymbol{a} = \boldsymbol{u}$ and $\nabla \cdot \boldsymbol{u} = 0$, (6) gives $\nabla \times \nabla \times \boldsymbol{u} = -\nabla^2 \boldsymbol{u}$, whose $y$-component is $-\nabla^2 v$. Hence

$$\big[\nabla \times \nabla \times \partial_t \boldsymbol{u}\big]_y = -\,\partial_t \nabla^2 v. \tag{7}$$

**Pressure term.** $\nabla \times \nabla \times (\nabla \Pi) = \mathbf{0}$ since the curl of a gradient vanishes. Pressure is eliminated.

**Viscous term.** $\nabla \times \nabla \times (\nabla^2 \boldsymbol{u}) = \nabla^2(\nabla \times \nabla \times \boldsymbol{u}) = \nabla^2(-\nabla^2 \boldsymbol{u}) = -\nabla^4 \boldsymbol{u}$, $y$-component $-\nabla^4 v$.

**Forcing term.** By (6), $\big[\nabla \times \nabla \times \boldsymbol{H}\big]_y = \partial_y(\nabla \cdot \boldsymbol{H}) - \nabla^2 H_y$.

Collecting,

$$-\,\partial_t \nabla^2 v = \partial_y(\nabla \cdot \boldsymbol{H}) - \nabla^2 H_y - \frac{1}{\mathrm{Re}}\,\nabla^4 v, \tag{8}$$

and multiplying by $-1$,

$$\boxed{\partial_t\, \nabla^2 v = \underbrace{\nabla^2 H_y - \partial_y(\nabla \cdot \boldsymbol{H})}_{h_v} + \frac{1}{\mathrm{Re}}\,\nabla^4 v} \tag{9}$$

The forcing simplifies because the $\partial_{yy} H_y$ pieces cancel:

$$h_v \;=\; \nabla^2 H_y - \partial_y(\nabla \cdot \boldsymbol{H}) \;=\; \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2}\right) H_y - \frac{\partial}{\partial y}\left(\frac{\partial H_x}{\partial x} + \frac{\partial H_z}{\partial z}\right). \tag{10}$$

## 3 The wall-normal vorticity equation

Apply $\big(\nabla \times \cdot\big)$ to (5) and take the $y$-component. The pressure term drops ($\nabla \times \nabla \Pi = \mathbf{0}$), $\big[\nabla \times \partial_t \boldsymbol{u}\big]_y = \partial_t \eta$, the viscous term gives $\frac{1}{\mathrm{Re}} \nabla^2 \eta$, and $\big[\nabla \times \boldsymbol{H}\big]_y = \partial_z H_x - \partial_x H_z$. Hence

$$\boxed{\partial_t \eta = \underbrace{\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x}}_{h_\eta} + \frac{1}{\mathrm{Re}}\,\nabla^2 \eta} \tag{11}$$

## 4 Fourier–Chebyshev representation

Expand every field in Fourier modes in the periodic directions and Chebyshev polynomials $T_n$ in $y$,

$$q(x, y, z, t) = \sum_{k_x} \sum_{k_z} \hat{q}(k_x, k_z, y, t)\, e^{\mathrm{i}(k_x x + k_z z)}, \qquad \hat{q}(\cdot, y, \cdot) = \sum_n a_n(t)\, T_n(y). \tag{12}$$

Then $\partial_x \to \mathrm{i}k_x$, $\partial_z \to \mathrm{i}k_z$, $\partial_y \to \mathcal{D}$ (Chebyshev differentiation), and with $k^2 \equiv k_x^2 + k_z^2$,

$$\nabla^2 \to \mathcal{D}^2 - k^2 \equiv L, \qquad \nabla^4 \to (\mathcal{D}^2 - k^2)^2 = L^2. \tag{13}$$

The evolution equations (9), (11) become, for each $(k_x, k_z)$,

$$\partial_t\, L\, \hat{v} = \hat{h}_v + \frac{1}{\mathrm{Re}}\, L^2\, \hat{v}, \qquad \hat{h}_v = -k^2\, \hat{H}_y - \mathrm{i}\,\mathcal{D}\Big(k_x \hat{H}_x + k_z \hat{H}_z\Big), \tag{14}$$

$$\partial_t\, \hat{\eta} = \hat{h}_\eta + \frac{1}{\mathrm{Re}}\, L\, \hat{\eta}, \qquad \hat{h}_\eta = \mathrm{i}\,\Big(k_z \hat{H}_x - k_x \hat{H}_z\Big). \tag{15}$$

(The factors $\mathrm{i}k_x, \mathrm{i}k_z$ commute with $\mathcal{D}$, so the $y$-derivative in $\hat{h}_v$ acts on $k_x \hat{H}_x + k_z \hat{H}_z$.)

## 5 Recovering $u$, $w$ and the vorticity

For $k^2 > 0$ the in-plane velocities follow from continuity (2) and the definition of $\eta$:

$$\mathrm{i}k_x\,\hat{u} + \mathcal{D}\hat{v} + \mathrm{i}k_z\,\hat{w} = 0, \tag{16}$$

$$\mathrm{i}k_z\,\hat{u} - \mathrm{i}k_x\,\hat{w} = \hat{\eta}. \tag{17}$$

This $2 \times 2$ system has determinant $k^2$, giving

$$\boxed{\hat{u} = \frac{\mathrm{i}}{k^2}\big(k_x\,\mathcal{D}\hat{v} - k_z\,\hat{\eta}\big), \qquad \hat{w} = \frac{\mathrm{i}}{k^2}\big(k_z\,\mathcal{D}\hat{v} + k_x\,\hat{\eta}\big)} \qquad (k^2 > 0). \tag{18}$$

The three vorticity components needed to build $\boldsymbol{H}$ are

$$\hat{\omega}_x = \mathcal{D}\hat{w} - \mathrm{i}k_z\,\hat{v}, \qquad \hat{\omega}_y = \hat{\eta}, \qquad \hat{\omega}_z = \mathrm{i}k_x\,\hat{v} - \mathcal{D}\hat{u}. \tag{19}$$

## 6 The mean mode $(k_x, k_z) = (0, 0)$

At $k = 0$ the recovery (18) is singular and is replaced by direct evolution of the plane-averaged profiles. Continuity and the no-penetration wall condition force $\hat{v}(0,0,y) \equiv 0$, and $\hat{\eta}(0,0,y) \equiv 0$ identically. The mean streamwise and spanwise profiles $u_0(y) \equiv \hat{u}(0,0,y)$, $w_0(y) \equiv \hat{w}(0,0,y)$ obey the $x$- and $z$-momentum equations at $k = 0$:

$$\partial_t u_0 = \hat{H}_x(0,0,y) + \frac{1}{\mathrm{Re}}\,\mathcal{D}^2 u_0 + f_x(t), \tag{20}$$

$$\partial_t w_0 = \hat{H}_z(0,0,y) + \frac{1}{\mathrm{Re}}\,\mathcal{D}^2 w_0, \tag{21}$$

where $f_x(t)$ is the spatially uniform mean pressure gradient. For a constant-mass-flux simulation $f_x(t)$ is chosen each step so that $\int_{-1}^{1} u_0\,dy$ equals its target; this is a single linear adjustment (superpose the unforced solution and the unit-forcing response).

## 7 Boundary conditions

No-slip / no-penetration at $y = \pm 1$ ($u = v = w = 0$) translates to

$$\hat{v} = 0, \quad \mathcal{D}\hat{v} = 0 \;\; \text{(from continuity with } \hat{u} = \hat{w} = 0), \qquad \hat{\eta} = 0, \qquad u_0 = w_0 = 0, \qquad \text{at } y = \pm 1. \tag{22}$$

Thus $\hat{v}$ carries *four* homogeneous conditions ($\hat{v} = \mathcal{D}\hat{v} = 0$ at each wall), while $\hat{\eta}$, $u_0$, $w_0$ each carry two (homogeneous Dirichlet).

## 8 Semi-implicit time discretization

Nonlinear terms are advanced explicitly with the low-storage three-stage Spalart–Moser–Rogers (SMR) scheme; the viscous terms are treated implicitly (Crank–Nicolson). For a generic field $\hat{q}$ obeying $\partial_t \hat{q} = N + \frac{1}{\mathrm{Re}} L\hat{q}$ (with $N$ the explicit forcing and $L = \mathcal{D}^2 - k^2$), each stage $k = 1, 2, 3$ advances $\hat{q}^{\,k-1} \to \hat{q}^{\,k}$ by solving

$$\Big[1 - \beta_k \tfrac{\Delta t}{\mathrm{Re}} L\Big]\hat{q}^{\,k} = \Big[1 + \alpha_k \tfrac{\Delta t}{\mathrm{Re}} L\Big]\hat{q}^{\,k-1} + \gamma_k\,\Delta t\,N^{k-1} + \zeta_k\,\Delta t\,N^{k-2}, \tag{23}$$

where $\hat{q}^{\,0}$ is the start-of-step value and $\hat{q}^{\,3}$ the end-of-step value. The two-register nonlinear combination $\gamma_k N^{k-1} + \zeta_k N^{k-2}$ needs only the current and previous stage forcings. The coefficients are

| stage $k$ | $\gamma_k$ | $\zeta_k$ | $\gamma_k + \zeta_k$ | (substep) |
|---|---|---|---|---|
| 1 | 8/15 | 0 | 8/15 | |
| 2 | 5/12 | −17/60 | 2/15 | |
| 3 | 3/4 | −5/12 | 1/3 | |

with the viscous split satisfying $\alpha_k + \beta_k = \gamma_k + \zeta_k$ (the three substep fractions sum to 1, a full step $\Delta t$). The symmetric Crank–Nicolson choice is $\alpha_k = \beta_k = \frac{1}{2}(\gamma_k + \zeta_k)$; SMR's stability-optimised variant uses $\alpha = (29/96, -3/40, 1/6)$, $\beta = (37/160, 5/24, 1/6)$. *The code implements the symmetric split* ($\beta_k = \frac{1}{2}C_k$ in `stepper_mod` and `par_stepper_mod`); because the symmetric Crank–Nicolson weights dominate the explicit RK3 side, the resulting scheme is **second order in time**, as measured directly by `test_stepper_order_r2c`.

Rearranging (23), the implicit operator is a **modified Helmholtz problem**

$$\Big[\mathcal{D}^2 - (k^2 + \lambda_k)\Big]\hat{q}^{\,k} = (\text{known RHS}), \qquad \lambda_k = \frac{\mathrm{Re}}{\beta_k\,\Delta t} \;=\; \frac{2\mathrm{Re}}{(\gamma_k + \zeta_k)\,\Delta t}\ (\text{symmetric CN}), \tag{24}$$

which for $\hat{\eta}$ and the mean profiles $u_0, w_0$ is solved with homogeneous Dirichlet data ($\hat{\eta} = 0$, $u_0 = w_0 = 0$ at $y = \pm 1$).

For $\hat{v}$ the time derivative acts on $L\hat{v}$, so (23) applied to (14) gives the **fourth-order (biharmonic-type) problem**

$$\Big[L - \beta_k \tfrac{\Delta t}{\mathrm{Re}}\, L^2\Big]\hat{v}^{\,k} = L\Big[1 + \alpha_k \tfrac{\Delta t}{\mathrm{Re}} L\Big]\hat{v}^{\,k-1} + \gamma_k\,\Delta t\,\hat{h}_v^{\,k-1} + \zeta_k\,\Delta t\,\hat{h}_v^{\,k-2}, \tag{25}$$

with the four conditions $\hat{v} = \mathcal{D}\hat{v} = 0$ at $y = \pm 1$. The operator $L - \beta_k \frac{\Delta t}{\mathrm{Re}} L^2$ is exactly the $(L - \alpha L^2)$ form assembled by the biharmonic routine, with $\alpha = \beta_k \Delta t/\mathrm{Re}$. The RHS requires two applications of $L = \mathcal{D}^2 - k^2$ to $\hat{v}^{\,k-1}$.

**Per-substep evaluation.** Within one RK substep the solver therefore (i) recovers $\hat{u}, \hat{w}$ and $\hat{\boldsymbol{\omega}}$ from $(\hat{v}, \hat{\eta}, u_0, w_0)$ via (18)–(19); (ii) transforms $u, v, w, \omega_x, \omega_y, \omega_z$ to physical space on the 3/2-dealiased grid; (iii) forms $\boldsymbol{H} = \boldsymbol{u} \times (\boldsymbol{\omega} + 2\boldsymbol{\Omega}) + \boldsymbol{F}$ pointwise,

$$H_x = v\,\omega_z - w\,\omega_y, \quad H_y = w\,\omega_x - u\,\omega_z, \quad H_z = u\,\omega_y - v\,\omega_x \quad (+\,2\boldsymbol{\Omega},\ \boldsymbol{F}\ \text{terms}); \tag{26}$$

(iv) transforms $H_x, H_y, H_z$ back to spectral space; (v) assembles $\hat{h}_v, \hat{h}_\eta$ via (14)–(15) and the mean $\hat{H}_x(0,0), \hat{H}_z(0,0)$; and (vi) performs the implicit solves above.

# A Pressure recovery for diagnostics (not used in time stepping)

Pressure is absent from the integration but may be reconstructed from a velocity snapshot. Taking the divergence of (5) and using $\nabla \cdot \boldsymbol{u} = 0$ (hence $\nabla \cdot \partial_t \boldsymbol{u} = 0$) gives a Poisson equation for the modified pressure,

$$\nabla^2 \Pi = \nabla \cdot \boldsymbol{H}, \qquad \text{i.e.} \qquad (\mathcal{D}^2 - k^2)\hat{\Pi} = \mathrm{i}k_x \hat{H}_x + \mathcal{D}\hat{H}_y + \mathrm{i}k_z \hat{H}_z. \tag{27}$$

The wall condition comes from the $y$-momentum equation evaluated at the wall, where $v \equiv 0$ for all time so $\partial_t v = 0$:

$$\mathcal{D}\hat{\Pi}\big|_{y=\pm 1} = \hat{H}_y + \frac{1}{\mathrm{Re}}\,\mathcal{D}^2\hat{v}\Big|_{y=\pm 1} \qquad (\textbf{Neumann}). \tag{28}$$

Finally $p = \Pi - \frac{1}{2}u_j u_j$. The $(0,0)$ mode is a pure Neumann problem, fixing $\hat{\Pi}$ only up to an additive constant (set, e.g., by zero mean pressure). Because the boundary data are Neumann, this requires a Helmholtz solver built on a Neumann (rather than Dirichlet) basis — a separate routine from the one used for $\hat{\eta}$.

# B Operator summary

| Field | Implicit operator | Boundary conditions |
|---|---|---|
| $\hat{\eta}$ | $L - \lambda$ (Helmholtz) | $\hat{\eta} = 0$ at $y = \pm 1$ |
| $u_0,\ w_0$ | $\mathcal{D}^2 - \lambda$ (Helmholtz, $k = 0$) | $u_0 = w_0 = 0$ |
| $\hat{v}$ | $L - \beta L^2$ (biharmonic) | $\hat{v} = \mathcal{D}\hat{v} = 0$ at $y = \pm 1$ |
| $\hat{\Pi}$ (diagnostic) | $L$ (Poisson) | $\mathcal{D}\hat{\Pi}$ prescribed (Neumann) |

with $L = \mathcal{D}^2 - k^2$, $\lambda_k = \mathrm{Re}/(\beta_k\,\Delta t)$ the Helmholtz shift and $\beta_k \Delta t/\mathrm{Re}$ the biharmonic factor (per stage; symmetric CN gives $\beta_k = \frac{1}{2}(\gamma_k + \zeta_k)$).